KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 1 REPORT

Team Formation & Data Hunting

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Lakshmi Prabha M M	25BAD049

The Cleaning Sprint – Data Quality Analysis

1. Week 3 Contribution

Studied the effects of missing values and outliers on statistical analysis and data distribution.

2. Work Done:

During Week 3, my individual work focused on understanding how missing values and outliers can affect the accuracy of statistical analysis in the Youth Smoking and Drug Consumption dataset. I researched different methods used to identify missing data and detect unusual observations. I also explored how these data quality issues can influence measures such as Mean, Median, Variance and Standard Deviation.

3. Topics Studied:

- **Missing Values – Studied how missing observations affect statistical calculations and reduce the reliability of data analysis.**
- **Missing Value Handling – Researched different imputation techniques, including Mean, Median and Mode imputation, and understood when dropping missing records may be appropriate.**
- **Outlier Detection – Studied the identification of unusually high or low values that may affect the overall distribution of numerical features.**

- **Interquartile Range (IQR) – Explored the IQR method for detecting outliers using the first quartile, third quartile and calculated lower and upper limits.**
- **Z-Score Method – Researched how Z-scores help identify observations that are far away from the mean of a distribution.**
- **Effect on Statistical Measures – Studied how extreme values can influence the Mean, Variance and Standard Deviation, while the Median is generally less sensitive to outliers.**
- **Data Distribution – Understood how missing values and outliers can affect the shape and interpretation of statistical distributions.**

4. Outcome of My Contribution:

This study helped me understand the importance of data quality before performing statistical analysis. I learned how missing values and outliers can influence statistical measures and how suitable handling techniques can improve the reliability of the analysis. This knowledge will be useful for interpreting the statistical results obtained from the cleaned dataset.

5. Reference:

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.